

Project report

What we have built, how we compare to the paper, and what to do next.

EXECUTIVE SUMMARY

Built: a production-shape product (server + UX + physics + eval + CI + docs) on top of a paper-faithful research model. Four genuinely above-the-paper additions shipped this session.

Vs paper: they win 8 dimensions (research substance), we win 11 (product + ops + extensions + hygiene), 2 different surfaces. Their trained-model quality is real; ours is smoke until M7 runs.

Next step: ONE cold email this week to an MLS analyst. Then \$30 for M7. Everything else is downstream of those two.

TESTS PASSING	LINES OF CODE	FILES WRITTEN	CUSTOMERS
30 / 30	~8,200	36	0

SECTION 1

What we have built

Concrete inventory of every component, what it does, and its line count. Excludes auto-installed skills, generated data, and the .venv. All paths relative to repo root.

MODEL CORE 1,452 LOC	INFERENCE SERVER 548 LOC	RENDERER 1,294 LOC	SCRIPTS 1,137 LOC
TESTS 316 LOC	DATA PIPELINE 269 LOC	LANDING SITE 1,977 LOC	DOCS 1,080 lines

Model & inference (src/model/)

FILE	LOC	WHAT IT DOES
config.py	86	Single source of truth for all hyperparams. Schema-versioned. Smoke + paper defaults.
tokenizer.py	116	Multi-agent trajectory tokenizer + <i>learned waypoint embedding</i> (above the paper).
backbone.py	108	TimeSformer-style factorized spatiotemporal attention (paper-faithful).
diffusion.py	259	DDPM noise schedule + ϵ -prediction network + synthetic-arrow CFG dropout training.
samplers.py	258	Reverse-diffusion + causal sliding-window rollout + classifier-free guidance + waypoint pinning.
physics.py	171	7-stage post-processor (seam, speed-clip, smooth, clamp, repulse, re-clip, re-clamp).
event_head.py	125	Hierarchical event classifier (paper-faithful; parked, labels not aligned).
dataset.py	157	Trajectory window Dataset reading converted GenTac JSON.
lightning_module.py	172	LightningModule + DataModule + EMA shadow weights + cfg roundtrip + schema-version refuse-on-load.

Inference server (src/server/main.py , 548 LOC)

CAPABILITY	DETAIL
✓ HTTP API	<code>POST /api/generate</code> + <code>GET /api/health</code>
✓ Streaming endpoint	<code>POST /api/generate/stream</code> — Server-Sent Events: 1 metadata + N window + 1 done (<i>this session</i>)
✓ Multi-arrow joint plausibility	Rejects <code>v_max-impossible</code> / <code>r_min-violating</code> pin sets with explicit reasons (<i>this session</i>)
✓ Concurrency control	Async semaphore (1 active gen) + queue cap (8) + 503 past that
✓ API-key auth	<code>X-API-Key</code> header, constant-time compare via <code>secrets.compare_digest</code>
✓ JSON-structured logs	Per-request <code>X-Request-Id</code> propagated request → response
✓ Path-traversal-safe match resolution	Anchored under <code>data/processed/</code> , <code>.resolve()</code> + <code>.relative_to()</code>
✓ Pydantic-bounded schema	$k \leq 32$, horizon ≤ 250 frames, ≤ 23 arrows, <code>guidance_scale</code> $\in [0, 5]$
✓ Discriminated Arrow union	<code>{kind:"player"}</code> + <code>{kind:"ball_pass", recipient}</code>
✓ Ball-pass receiving-stance offset	Recipient lands 1.50m perpendicular to approach, not on the ball
✓ CORS allowlist	Configurable via <code>GENTAC_CORS_ORIGINS</code> env var

Renderer / coach UX (`src/render/`, `index.html`)

FEATURE	LOC	DETAIL
✓ Tactics board canvas	pitch.js: 919 style.css: 291 index.html: 84	FIFA-spec markings, 1050×680, trails, scrubber, play/pause, speed control
✓ Click-drag player arrows (killer mechanic)		Hover ring, in-progress preview, right-click clear
✓ Ball arrows + recipient designation		Drag ball, click recipient — pulsing yellow rings
✓ Touch event normalization		iPad/tablet works without separate code path
✓ Compare mode + overlay-diff		Ghosts actual onto alternative panel
✓ First-run onboarding overlay		3-step instructions, dismissible, localStorage-persisted

Eval harness (`scripts/eval.py` , 474 LOC)

METRIC	DETAIL
✓ ADE / FDE / diversity / off-pitch / max-speed	All with 95% percentile bootstrap CIs (1,000 resamples default)
✓ <code>--compare A B</code>	Paired bootstrap p-values, significance markers ($*** < 0.01$)
✓ Per-role ADE	GK / defender / midfielder / forward / ball — <i>above the paper (this session)</i>
✓ Realism discriminator	Small MLP, real vs generated; test acc 0.5 = indistinguishable — <i>above the paper (this session)</i>
✓ CFG-scale sweep	Arrow-honoring err + hit-rate across guidance scales
✓ Reproducible	<code>--seed</code> for full determinism

Training, data, & cloud pipeline

FILE	LOC	WHAT IT DOES
<code>scripts/smoke_train.py</code>	62	3-epoch MPS smoke; <code>pl.seed_everything</code> for determinism. ~80 s.
<code>scripts/train_full.py</code>	141	Paper config + <code>--dry-run</code> mode (1 batch on CPU, validates entire cloud code path) + <code>--seed</code> + <code>--ema-decay</code> .
<code>scripts/modal_train.py</code>	253	Modal entrypoints + lazy import + <code>--validate-local</code> (works without modal installed).
<code>scripts/sample.py</code>	207	K-sample inference + physics + JSON dump.
<code>src/data/metrica_to_gentac.py</code>	218	Metrica CSV → GenTac JSON + strict <code>validate_payload()</code> .
<code>src/data/extract_clip.py</code>	51	Slice 30 s dev clips + warning-mode validator.

Tests & CI

FILE	LOC	COVERAGE
<code>tests/test_physics.py</code>	70	Speed-clip invariant, on-pitch clamp under adversarial input, seam anchor, no-NaN
<code>tests/test_config_schema.py</code>	68	cfg dict roundtrip, schema-version refuse-on-load, smoke vs paper config
<code>tests/test_server_validation.py</code>	129	12 Pydantic cases (arrow union, bounds) + 4 path-traversal cases
<code>tests/test_eval_helpers.py</code>	43	Bootstrap CI brackets the mean, paired p-value sanity, identical-lists handling
<code>.github/workflows/tests.yml</code>	40	CPU torch + lightning install, fetch Metrica, run pytest on every push/PR

Verified end-to-end this session: 30 passed in 3.16s .

Landing site (`site/` , 1,977 LOC)

FILE	LOC	DETAIL
index.html	433	Dark editorial, premium typography, full league-pitch structure
style.css	1005	Design system, scroll-driven animations, responsive, prefers-reduced-motion respected
app.js	511	Animated canvas (22 reactive players + ball-pass scenarios), scroll progress, reveal animations, contact form
vercel.json	28	Security headers + cache control + clean URLs
public/og.svg	—	1200×630 social-share image, branded
public/favicon.svg	—	Inline SVG "B" mark in brand orange

Documentation ([docs/](#) + [README.md](#) , 1,080 lines)

DOC	LINES	PURPOSE
README.md	319	Buyer-facing 60-second preamble + developer onboarding + roadmap
docs/handoff_api.md	330	v0.2 spec: discriminated arrow union + Python SDK + vendor compatibility matrix
docs/cloud_training.md	172	Pre-flight + 5-section troubleshooting matrix
docs/operations.md	162	Production topology + cost envelope + observability gaps + security checklist
docs/mvp.md	97	One persona, one moment, IN/OUT scope, four validation questions

Comparison to the paper

Paper: Rao, Gui, Wu, Wang, Xie — GenTac: Generative Modeling and Forecasting of Soccer Tactics, arXiv:2604.11786, 13 Apr 2026 (SJTU + Nanjing). Verified against the extracted text of [gentac.pdf](#). Our figures from this session's eval against the smoke checkpoint.

Where the paper wins (substance gaps)

#	ASPECT	PAPER	US
1	Original architecture	Novel publication	Faithful reimpl (parity at best)
2	Training scale	60 epochs × A100, batch 200 §6.3.3	3 epochs × MPS, batch 4 (smoke); cloud script dry-run-validated
3	Training data	TacBench: 2,838 × 15s segments, multi-league §1.250	~7,200 windows from 2 Metrica open-sample matches
4	Min ADE @ 1s (best of K=20)		0.62 m §3.1 1.310 n/a (we report mean, not min)
5	Mean ADE @ 1s	≈1.0–1.2 m (inferred from §1.311)	3.50 m [95% CI 2.63–4.13] — ~3× worse
6	Min FDE @ 5s (opp-conditioned)		2.89 m §3.1 1.326 n/a — we only evaluate @ 1 s
7	Style conditioning (team/league/objective)	A-League cond. cuts ADE 38% at 1s §3.1 1.355	None
8	Event recognition (15 classes)		F1 = 73.4% §3.2 1.468 Code complete, parked (labels not aligned)
9	Peer review / citability	Published arXiv, formal refs	Engineering prototype

Where we win (product + ops + extensions)

#	ASPECT	PAPER	US
10	Physics post-processor	None (raw model output)	7-stage; 0% off-pitch, vmax exactly 10.5 m/s
11	Inference server	None published	FastAPI + auth + queue + JSON logs + req IDs
12	Streaming inference (SSE)	N/A	Implemented this session
13	Multi-arrow joint plausibility	Paper has no arrow input	Server-side check with explicit reasons (this session)
14	Coach UX (player + ball arrows, touch, overlay)	None — research code only	Full implementation + first-run onboarding
15	Eval methodology	Min/mean over K=20, multi-horizon ablations	95% bootstrap CIs + paired bootstrap p-values + --compare
16	Per-role ADE breakdown	Aggregate only	GK 3.60m / def 3.54m / mid 3.53m / fwd 3.34m / ball 0.69m (this session)
17	Realism discriminator metric	Not reported	MLP test acc 0.667 — 0.50 indistinguishable, 1.00 trivially separable (this session)
18	Cloud-training automation	N/A	--dry-run + --validate-local + troubleshooting matrix
19	Handoff to downstream video gen	Not in scope	Schema v0.2 + Python SDK + vendor compat notes
20	Tests / CI	N/A	30 tests passing, GitHub Actions on every push/PR

Different surfaces (neither strictly better)

#	ASPECT	PAPER FRAMING	OUR FRAMING
21	Counterfactual conditioning	Offensive/defensive objective conditioning alters threat metrics §3.1	Arrow + CFG (coach gesture as the input modality)
22	Architectural extensions	Style/team/league/objective + event head	Learned waypoint + CFG + EMA + ball-pass with receiving offset

Honest summary. The paper is a stronger **trained model and benchmark**. We are a stronger **product around that model**. The composition would work cleanly: take the paper's trained weights, drop them into our inference server + renderer + handoff spec, and you have a deployable product backed by paper-grade model quality. The single missing piece in our column is the trained model itself — a ~\$30 cloud run we have scripted and validated locally but have not executed because we lack the real league tracking data that would justify the GPU time. That gap is commercial, not technical.

SECTION 3

Next steps

Priority-ordered. Capital and time estimates assume current Mac-laptop development environment and no co-founder. Adjust upward if scope or quality bar grows.

This week — zero capital required

#	ACTION	WHY IT MATTERS
1	Send 5 cold emails to MLS analyst-level contacts. LinkedIn search "MLS data scientist" / "performance analyst." Subject line with the word "arrows." Ask, don't pitch.	Closes the M0 gap that has not moved in 10+ sessions. Single highest-leverage action.
2	Email SkillCorner + Stats Perform asking about startup data eval programs.	3–6 month sales cycle starts now or it doesn't start.
3	Pick a real brand name that isn't "B." Update the landing page.	Single-letter brand without founder equity signals "I haven't thought about brand."
4	Soften any remaining over-claims on the landing page.	Trust is burned in 30 seconds if a buyer clicks through and discovers fiction.

Success metric for the week: 1 cold-email reply, 1 data-partner reply.

This month — ~\$50 capital

#	ACTION	WHY IT MATTERS
5	Run M7 cloud training (\$25–30 on Lambda Labs A100). Code is dry-run-validated; troubleshooting matrix exists.	Moves rows 1, 3, 5, 6, 16, 17 in our favor simultaneously. Realism disc should drop from 0.67 toward 0.50; mean ADE @ 1s should drop from 3.50 m toward 1.0–1.5 m.
6	Land 1 evaluation agreement with an analyst from cold-email replies.	First data point on whether anyone actually wants this.
7	Pick a 3D video vendor and ship a one-clip integration.	Without this, the demo is dots on a green pitch — not what the pitch sells.
8	Re-record the landing-page demo video with the M7-trained model.	Current canvas demo is heuristic; real model output is the differentiator.

Success metric for the month: signed evaluation agreement + working M7 model + 3D clip in the demo.

This quarter — ~\$10K capital + 1 co-founder

#	ACTION	WHY IT MATTERS
9	Recruit a co-founder with sports-industry pedigree. 30–40% equity.	Without this, no institutional investor takes a meeting. Sports tech is the most relationship-driven enterprise market.
10	Sign a real evaluation-tier data deal (SkillCorner most likely path; \$0–\$10K/year).	Real data is what makes M7 actually translate to product value.
11	Deploy to production for first design partner. Topology in docs/operations.md .	Single A10G VM + Caddy + S3; ~\$250–\$450/month per single-tenant customer.
12	Raise \$250K friends-and-family round.	6 months of runway for the founder + travel for league demos. Pre-product seed rounds are realistic at this size; \$500K+ requires the co-founder + LOI from step 9.

Success metric for the quarter: co-founder onboarded + first paid pilot (\$10K–\$50K) signed.

The one thing that has not changed. Every code addition, every score improvement, every paper-comparison win we have shipped in this session and prior — none of it moves M0. **M0 moves when an email is sent.** No code has ever been sent in an email. If the next session is also engineering work, the honest interpretation is either "this is a portfolio project" (fine — ship open-source) or "you are avoiding the sales work" (which kills the company). There is no third interpretation consistent with another quarter of pure tech work.

Summary table

HORIZON	CAPITAL	PEOPLE	OUTCOME TARGET
This week	\$0	Just you	1 cold-email reply
This month	~\$50	Just you	Signed evaluation + M7 model
This quarter	~\$10K + \$250K F&F	+1 co-founder	First paid pilot
Months 6–12	\$500K–\$900K seed	+1 ML eng, +1 BD	2nd paying customer, SOC 2 Type I